• As to the terms with serial numbers from 1 to N, it is always possible to select both the greatest (denoted by B_1) and the least (denoted by A_1) terms since the number of these terms is $\boxed{\text{finite}}$

↳ Needs Justification (San)

• Now we have to select the least value from $a-\varepsilon$ and A_1 (denoted by A) and the greatest value from $a+\varepsilon$ and B_1 (denoted by B). It is obvious that $A \leq y_n \leq B$ for all the terms of our sequence, which proves that the sequence (y_n) is bounded.

↳ Sena conclusion. Darun proof

Reader: I see

Author: Not too well, it seems. Let us have a look at the logical structure of the proof. We must verify that if the sequence has a limit, there exist two numbers A and B such that $A \leq y_n \leq B$ for each term of the sequence. Should the sequence contain a $\boxed{\text{finite}}$ number of terms, the existence of such two numbers would be evident. However, the sequence contains an $\boxed{\text{infinite}}$ number of terms, the fact that complicates the situation.

Reader: Now it is clear! The point is that if a sequence has a limit a, one concludes that in the interval from $a-\varepsilon$ to $a+\varepsilon$ we have an infinite set of y_n starting from $n=N+1$ so that $\boxed{\text{outside of this interval}}$ we shall find only a $\boxed{\text{finite}}$ number of terms (not larger than N).

Author: Quite correct. As you see, the limit "takes care of" all the complications associated with the behaviour of the infinite "tail" of a sequence. Indeed, $|y_n-a| < \varepsilon$ $\boxed{\text{for all } n > N}$, and this is the main "delicate" point of this theorem. As to the first N terms of a sequence, it is essential that their set is finite.

Reader: Now it is all quite lucid. But what about ε? It's value is not preset, we have to select it.

Author: A selection of a value for ε affects only N. If you take a smaller ε, you will get, generally speaking, a larger N. However, the number of the terms of a sequence which do not satisfy $|y_n-a| < \varepsilon$ will remain finite.

And now try to answer the question about the validity of the converse theorem: If a sequence is bounded, does it imply it is convergent as well?